NIHR | National Institute for Health and Care Research

Reference number	45512
Programme	Invention for Innovation
Call	i4i Call 29 Stage 1
Lead Applicant	Mr Ahmed Ahmed
Host organisation	Imperial College of Science, Technology and Medicine
Research Title	Side-to-Side Compression Anastomosis using the GT Metabolic Solutions Magnet System to Achieve Duodeno-Ileostomy bypass in Adults with Obesity and Type 2 Diabetes Mellitus.
Start Date	01/11/2025
Grant Duration (Months)	36
End Date	
Estimated Research Cost	£550,438.75
Estimated NHS Support & Treatment costs	£141,425.21
Research Type	Primary Research

Pre-submission

Lead Applicant Details

Please select if your application is a part of the following highlight notices or rolling calls:

Lead Applicant

Full Name	Mr Ahmed Ahmed	Institution	Imperial College of Science, Technology and Medicine
ORCID iD	0000-0002-8349-4688	Position	Clinical Reader in Metabolic Surgery

Specify your (lead applicant) role in the research
Mr A R Ahmed (Chief Investigator) is an expert in the clinical management of obesity and experienced clinical trialist. He was awarded in 2016 'Leading Commercial Principal Investigator' award by the NIHR. As chief investigator, he received £1,085,224.58 NIHR EME award (ID 17/1/49) in 2020 for the EMBIO trial: Left gastric artery embolisation for weight loss in obese patients with BMI 35-50kg/m2. As Lead Applicant, my role will include implementation and coordination of this study (including performing of interventions), recruitment of participants, data interpretation and final write-up of findings.

% FTE	10

Research Team Details

Co-Applicant	Organisation	Specify role in research	% FTE
Dr Melody Ni	Imperial College of Science, Technology and Medicine	Dr Ni will carry out staged and interrelated activities to understand health economic implications of using Magnet system in the NHS and potential barriers adoption.	50
Professor Alexander Miras	University of Ulster	Dr Miras will lead on analysis of type 2 diabetes remission following intestinal bypass intervention using the magnet system	2
Professor Toby Richards	University College London	Prof Richards has experience in clinical trials recently NIHR HTA PREVENTT) and will continue to advise on planning, design and conduct of the study	2
Ms Beata Szydlik	Cromwell Hospital	Ms Szydlik will is the trial PPIE Lead and will continue to lead on PPIE work-package for this trial.	1

COI declaration
No COI relating to this application

Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

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Obesity and Type 2 diabetes (T2DM) affect nearly 10% of the UK population leading to ill health costing the NHS significant amounts [1,2]. Bariatric (weight-loss) surgery is well established and a proven treatment for effective weight loss and T2DM. The most common procedures work by 'bypassing' parts of the gut. These operations are routine in the NHS and are recommended by many health advisory groups including NICE[3,4,5] .

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We know bariatric surgery produces good weight loss with excellent long-term reduction in T2DM [6,7] . However, access to bariatric surgery has been impacted by NHS pressures and of the 3.6 million people in the UK[8] who meet the criteria for bariatric surgery, only 4800 underwent surgery between 2022-2023[9].

The MAGNET System is a new medical device which can provide a solution to this. It uses 2 magnets, which are introduced through the mouth then positioned in a way that they join across 2 sections of bowel creating a "bypass". This removes the need to surgically divide the bowel and join it back together. The MAGNET System provides a simpler quicker and cheaper way to perform bariatric surgery. Early trials of the device (NCT05322122 and NCT05692518 ClinicalTrials.gov) indicate the MAGNET System to work, be safe and can be undertaken as a day case procedure.

We undertook a patient forum, involving 32 participants at Imperial College NHS Trust to explore if this may be something that would appeal and ask what factors would help guide patient interest. Quicker recovery time, sustainable weight loss and reduction in diabetes were key features of interest to patients.

In the first UK study of this device, Imperial College NHS Trust in collaboration with GT Metabolic Solutions Inc. will enrol 50 patients to see how well the MAGNET System works for weight loss and diabetes management. The study aims to provide a real-world evaluation to accelerate uptake of this product, which has already demonstrated safety and efficacy abroad. Furthermore, this trial would provide evidence which could lead to UK regulatory approval and use of the device in the UK, aligning with the NHS's goal to improve access to obesity and diabetes treatments.

This research has the potential to treat obesity by offering a safer, less invasive, and more affordable option, thereby allowing more patients to benefit from this treatment.

Research Plan

Research Plan

1. What are the problems being addressed?

Type 2 Diabetes Mellitus (T2DM) and Obesity:

By 2030, it is predicted that 10% of UK adults will have T2DM, with a third of these cases directly linked to obesity[1]. Both T2DM and obesity are associated with increased mortality, morbidity, and healthcare costs[2]. Despite pharmaceutical and lifestyle interventions, many patients with T2DM simply do not achieve treatment goals and remain at high risk of micro- and macrovascular complications. The UK Department of Health emphasises the need for more effective T2DM prevention and treatment strategies. Bariatric surgery, recommended by NICE[4], has shown superior long term glycaemic control with less diabetes medication use and higher rates of diabetes remission compared to medical/lifestyle interventions, according to pooled analysis of randomised controlled trials[10] and Cochrane reviews[11,12]. Current barriers to bariatric surgery include availability, accessibility, cost, eligibility, concerns regarding surgical complications, and social stigma[13]. One solution may be the development of novel technologies. The proposed trial uses magnets to perform bariatric surgery in an effective, safer, faster, and more acceptable and scalable manner.

Surgery:

Bariatric operations work by reducing food intake permanently through reducing stomach size, and/or re-routing (bypassing) food through the small intestine to reduce calorie absorption. Studies show that bypass surgery, as opposite to gastric reduction, is more effective in the management of T2DM. These procedures are safe, effective, and significantly improve HbA1c levels with a minimum of 50% resolution of diabetes in patient groups[14]. The National Obesity Audit of 2022/2023[9] reveals that over 50% of bariatric procedures in England are bypass surgeries.

Complications of bypass surgery mainly arise from the surgical anastomosis (join made between 2 intestine segments). Traditional methods rely on sutures or staples, which require an incision to the intestine (enterotomy) and leave behind foreign materials (i.e. staples or non-resorbable sutures). They are associated with complications such as bleeding (0.1–5%), leakage

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(0.1-8.3%), obstruction, and infection[15,16]. Furthermore anastomoses (during bariatric surgery) take time for the surgeon to perform (average 60 minutes)[17] and are expensive when stapling devices are used. The average length of stay following a bariatric bypass procedure is 2 days[18]. The MAGNET System addresses this challenge by using "compression anastomosis technology" to simplify the creation of anastomoses, traditionally the most challenging part of the procedure.

The MAGNET System uses magnets which are introduced orally. The magnets are then positioned laparoscopically at the duodena-ileal site. It works by compressing the tissue between the magnets, resulting in tissue necrosis and the formation of a side-to-side duodeno-ileal anastomosis (Figure 2 and 3). The magnets are then passed spontaneously in the patient's stool. Compression anastomosis technology has the potential be as safe effective, more time efficient but less expensive compared to current standard of care. In essence the magnet system mimics the anatomical design of the surgery but is much less invasive.

UK Regulatory Approval:

The MAGNET System is a product developed by GT Metabolic Solutions, Inc. The magnet dimensions are 39mm(L) x 10.2mm(W) x 4.4mm thickness and is composed of polyglycolic-co-lactic acid and barium sulfate. This device is the first magnetic compression anastomosis product intended for commercialisation by the company. It has US FDA Class II medical devices approval. To date, there have been 125 minimally invasive magnetic surgeries completed in clinical trials for side-to-side magnetic compression anastomosis. The MAGNET System was 510(k) cleared by the FDA in October 2024 and launched commercially in November in the USA. It is not commercially available in any other region. We propose an industry-academia collaboration to carry out the first UK clinical trial in order to gather vital UK-based evidence on safety and efficacy as a first step towards UK regulatory approval.

2. Why is this research important in terms of improving the health and/or wellbeing of the public and/or to patients and health care services?

In the UK 4.3 million people have diabetes[19]. Currently 90% of adults with T2DM are overweight or obese. One of the most modifiable risk factors for T2DM is obesity.

Bariatric small intestinal bypass procedures have been demonstrated to improve metabolic outcomes in adults with obesity. These procedures involve side-by-side anastomoses currently created by sutures or staples. The MAGNET System has the opportunity to provide patients living with T2DM complicated by obesity with a potentially safer and more cost-effective option than conventional anastomosis techniques thereby making the procedures more scalable. More specifically we expect using the magnets for creation of the anastomosis will (i) eliminate the potential surgical complications of anastomotic stricture, bleed (0-5-1%) or leak (0-8-3%). Over the course of the initial 6 US clinical studies, 0.0% of patients have sustained postoperative stricture, leakage or bleeding at 6- and 12-month follow-up using the magnet system[20-25]; (ii) reduce total operative time from 100 mins to 50 mins and (iii) reduce length of hospital stay from an average of 2 days[18] to same day discharge.

We discussed this novel technique with 32 members of our patient and public involvement and engagement (PPIE) group at Imperial College NHS Trust during a 2-hour meeting. They all expressed an interest to participate in this research study. Many people felt this new technology would be beneficial with regards to shorter procedure time, quicker recovery, less complications and provide access to more patients. The success rate demonstrated in the clinical trials abroad was encouraging to them and a good measure of success. The feedback and input we received from the PPIE group was invaluable as it shaped the design of the trial.

3. Review of existing evidence – How does the existing literature support this proposal?

The MAGNET System was evaluated in a prospective multi-center clinical study across Belgium, Spain, Canada, and Georgia, involving 49 patients over 12 months (NCT05322122). The participants were obese adults aged 18-65. The device was successfully placed in all subjects and all patients were discharged within 24 hours, with confirmed patent anastomosis in barium studies. All subjects passed the device naturally without needing re-intervention. Interim results have been reported in scientific literature as described below[26].

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Gagner[26] recently reported on magnet devices used in multi-center studies for creating side-to-side compression anastomosis duodeno-ileostomies in 24 patients. The devices were endoscopically delivered to the duodenum and ileum, successfully placed and passed naturally without surgical intervention, and expelled within a median of 48.5 days. There were no device-related adverse events, although three serious procedure-related adverse events occurred: a urinary tract infection, jejunal obstruction from an internal hernia, and a pelvic fluid collection. There were no instances of anastomotic bleeding, leaks, strictures, or mortality.

Based on data from the first-generation all-metal device, GT Metabolic developed the second-generation biofragmentable magnet device to improve dislodgement and ensure expulsion within 30 days, while maintaining anastomosis patency. This revised device was clinically tested in nine participants from December 2022 (NCT05692518), with all participants successfully completing a three-month follow-up. Both magnets were placed successfully, with the first swallowed and the second placed endoscopically. The median expulsion time for this group was 24 days, with expulsion and anastomosis patency confirmed radiologically. The upcoming study will employ this second-generation MAGNET System.

4. What is the research question/ aims and objectives?

The overall objective of this study is to evaluate the feasibility, performance, safety and efficacy of the MAGNET System to create side-to-side anastomoses to achieve duodeno-ileal bypass (Figure 1) in the NHS.

The study specific objectives and endpoints are defined in **Table 1**.

Table 1: Study Objectives and Endpoints Objective	**Endpoint**	**Justification of Endpoint**
Primary		
To assess the feasibility / performance of side-to-side anastomosis for duodeno-ileal diversion using the MAGNET System	Feasibility/performance at Day 90. The side-to-side anastomosis for duodeno-ileal diversion will be considered feasible if it results in successful: • Successful placement of the magnets (≥ 90% alignment of magnets); and • Passage of magnets without surgical re-intervention; and • Creation of a patent anastomosis confirmed radiologically. The primary endpoint will be met if the feasibility/performance is confirmed in >80% of enrolled and treated participants.	Feasibility / performance of the MAGNET System creation of a patent anastomosis is an appropriate measure of technical success.
Safety		
To evaluate the safety of the MAGNET System	Safety defined as incidence of serious adverse events related to the study device or study device procedure requiring additional emergency surgery or re-intervention, including: • All-cause mortality • Intestinal perforation and/or peritonitis • Intestinal obstruction • Life-threatening bleeding • Incidence of device malfunctions	The incidence of treatment emergent AEs and SAEs is a standard measure of the safety. The incidence of device malfunction informs about device reliability.

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Secondary Endpoints		
To provide information on the therapeutic effect of the side-to-side anastomosis for duodeno-ileal diversion using the MAGNET System	Efficacy will be assessed at Day 90, 180 and 360 based on change from baseline in: • anthropometrics (weight, BMI, %EWL) • metabolic markers (HbA1c, fating plasma glucose, number of medications for T2DM)	These are clinically relevant and validated measures of the therapeutic effect on weight loss, metabolic indicators measures, and patient-reported outcomes.
Health economic data	• Length of surgery and length of stay • Comparison of cost for surgery using MAGNET System vs conventional method • Cost of complications related to MAGNET System device	These are relevant in us assessing whether the MAGNET system is economically viable in the NHS
Patient and clinician satisfaction data	• Patient satisfaction questionnaire • Clinician satisfaction questionnaire	These are relevant in us assessing whether the MAGNET system is an acceptable method from a patient and clinician perspective

5. Project Plan

Overall Design:

Prospective, single arm, single centre clinical trial of the MAGNET System for creation of side-to-side anastomoses in eligible participants requiring a small intestinal bypass (duodeno-ileal diversion).

Total Expected Duration of the Study:

21 months (9 months recruitment and 12 month follow up)

Participants will have follow-up visits at 1, 14, 30, 90, 180, 270, and 360 days. When all enrolled participants have been followed for 12 months, or have exited the study, the study will be closed.

Number of Participants to be Enrolled:

This study will enroll up to 50 participants at Imperial College NHS Trust (ICHT). ICHT carries out approximately 400 bariatric bypass surgeries per year and therefore will be able to recruit and carry out 50 procedures in the study timeframe. Furthermore, the sample size has been chosen given the single arm design and feasibility study purpose.

Inclusion Criteria:

1.
    a. 18-65 years of age, inclusive, at the time of informed consent meeting NICE criteria for bariatric surgery
    b. BMI 30-50, inclusive with T2DM (defined as HbA1c > 6.5%)
    c. If a child-bearing female, subject must commit to not becoming pregnant and agree to use contraception for1 year
    d. Willing and able to comply with protocol requirements

Exclusion Criteria:

1. Use of injectable insulin
2. Use of GLP-1 analogues in last 2 months
3. Prior intestinal surgery
4. Implantable pacemaker or defibrillator

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5. Any contraindication to bariatric surgery
6. Expected need for MR imaging within the first 2 months after the procedure

Trial Procedure (Figure 1):

The procedure (Figure 3) is an inpatient procedure, taking place in an operating room under general anaesthesia. Each magnet is introduced orally either by swallowing [first distal magnet only] or with a flexible endoscopic catheter [second proximal magnet]. Final positioning of themagnets at the intended anastomosis site is completed using the magnetic Laparoscopic Positioning Device and standard laparoscopic instruments, to manipulate the magnet to the intended positions and lift the distal magnet in the ileum to meet with the proximal magnet in the duodenum. During the procedure, participants will be continuously monitored as per ICHT standard of care.

Upon completing the study procedure, any adverse effects, device deficiencies, protocol deviations will be recorded.

The study participants are expected to be discharged from the hospital the same day following the procedure, unless a longer stay is medically required as determined by the investigator. The successful placement of the MAGNET System will be confirmed radiologically (abdominal x-ray) and fluoroscopically (barium or gastrographin).

The participants are followed up at 14, 30, 60, 90, 180, 270 and 360 days.

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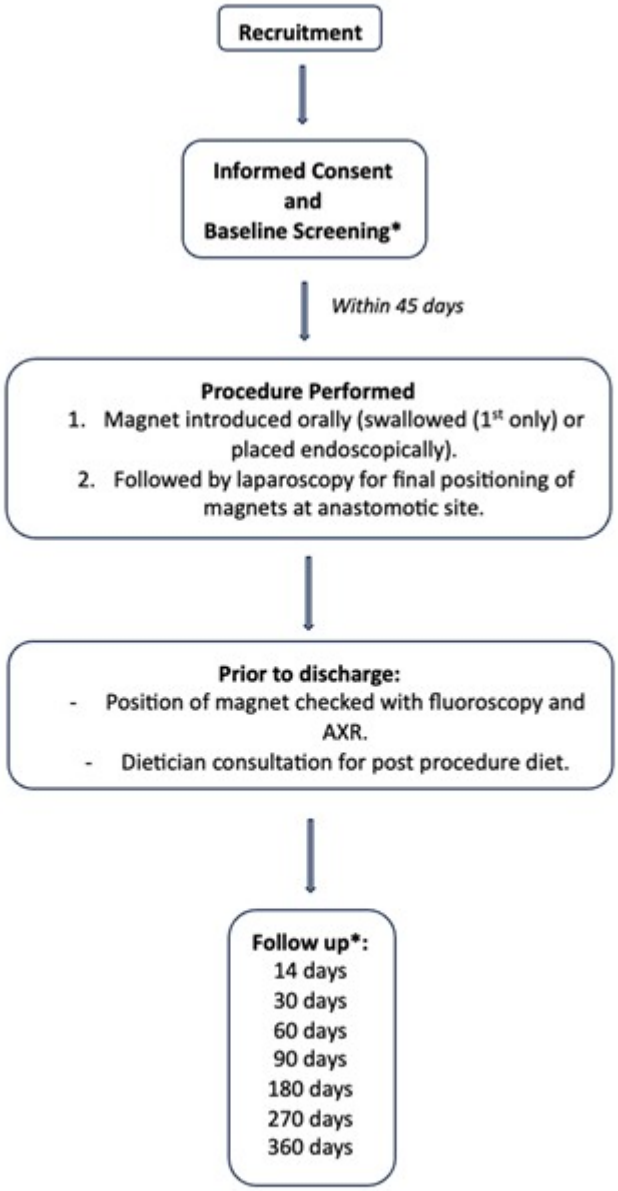

***Please see schedule of study assessments table**

Figure 1: Study Procedure Flow Chart

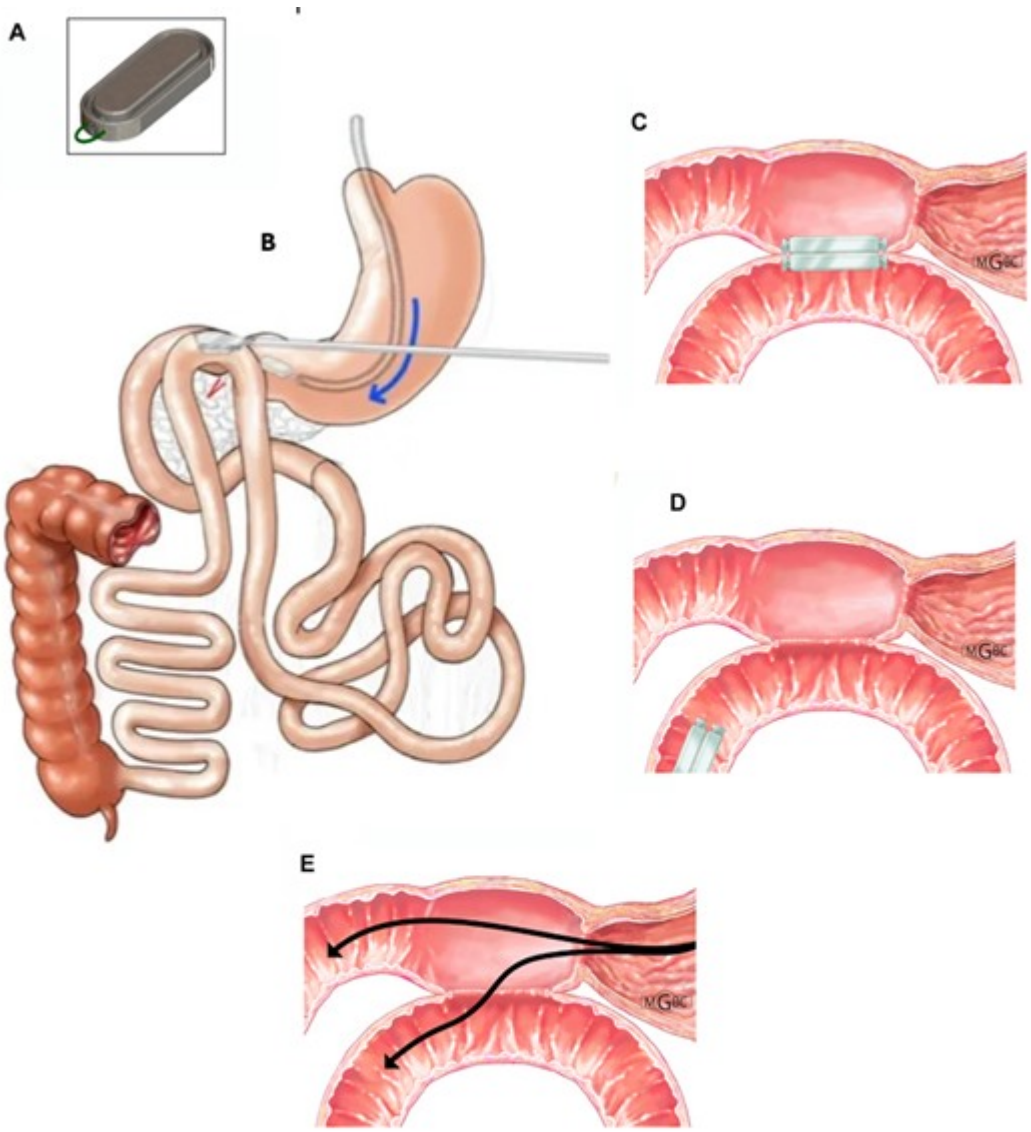

Figure 2: (A) Magnet Anastomosis System (MS) magnet. (B) After the distal MS magnet is endoscopically positioned at the ligament of Treitz and directed through the jejunal lumen to the ileum by laparoscopic magnetic positioning device assistance, it is raised anterior and latero-lateral to the post pyloric duodenum to align with the endoscopically delivered proximal MS magnet and the mesenteric space is closed. (C) In approximately 2–4 weeks, magnets fuse and the patent

Step 1

Patient swallows 1st magnet

1st magnet reaches ligament of Trietz

Step 2

Using a laparoscopic positioning device, the 1st magnet is move distally into the last 250-300cm of the ileum

Step 3

2nd magnet is placed in proximal duodenum using gastroscopy

Step 4

The 2 magnets are placed together, and mesenteric defect closed

Step 5 (Over 2-3 weeks)

Fusion of magnets resulting in tissue necrosis and creation of side-to-side anastomosis

Figure 3: Procedure: Side-to-Side Compression Anastomosis using the GT Metabolic Solution Magnet System to Achieve Duodeno-Ileostomy Diversion

ssion

Assessment	Informed Consent & Screening (Baseline)	Procedure (≤ 45d post-ICF)	(Discharge) (+ 4d)	Day 14 (Post-Op) (± 3d)	Day 30 (± 5d)	Day 60 (± 14d)	Day 90 (± 14d)	Day 180 (± 14d)	Day 270 (± 14d)	Day 360 (± 14d)
Informed Consent	X									
Inclusion / Exclusion Screening	X									
Demographics and Medical History	X									
Physical Exam Including Vital signs	X	X	X	X	X	X	X	X	X	X
Weight, Waist Circumference	X	X			X	X	X	X	X	X
Concomitant Medications	X	X	X	X	X	X	X	X	X	X
Psychologic Evaluation	X									
Surgical Clearance	X									
Clinical Laboratory Tests										
Pregnancy Test	X	X								
Blood Tests[1]	X				X	X	X	X	X	X
Imaging / Diagnostics										
Abdominal X-Ray (devices position/passing)		X[2]	X	X[3]	X[3]	X[3]	X[3]			
Fluoroscopy UGI (barium or gastrographin)			X		X		X			
Gastroduodenoscopy / Endoscopy		X			X[4]	X[4]	X[4]	X		X
Questionnaires (Surveys)										
Satisfaction Questionnaire								X		X
Other										
Device Deficiency Reporting		X	X	X	X	X				
Protocol Deviation Reporting	X	X	X	X	X	X	X	X	X	X
Adverse Event Reporting	X	X	X	X	X	X	X	X	X	X

Abbreviations: CBC, complete blood count; d, day; HbA1c, hemoglobin A1c; IC, Informed Consent; T2DM, type 2 diabetes mellitus.

[1] Blood tests include: CBC, coagulation profile *at screening*, lipid panel, liver profile, chemistry profile, thyroid function tests, HbA1c, vitamins and minerals (see **Section 6.2**.)

[2] Only for participants who opt to swallow distal Magnet (first, 1 of 2 Magnets).

[3] If Magnet passed previously, no further abdominal x-rays. Perform minimally at Day 30, and/or per PI discretion (as UVs if needed), until Magnet is no longer detectable.

[4] Gastroduodenoscopy/endoscopy performed the earlier of visit Day 30, 60, 90 following either Magnet anastomosis release (a) confirmed on x-ray, or (b) expelled/collected.

Table 2: Schedule of Study Assessments

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Statistical Methods:

The primary feasibility / performance endpoint and all secondary efficacy endpoints (safety evaluation and initial efficacy) identified here are considered a priori analyses because they have been pre-specified prior to initiating enrolment. All other, analyses, if any, designed subsequent to locking the database, will be considered post-hoc analyses, and their results will be considered exploratory. Any post hoc analyses will be clearly identified in the clinical study report.

Since this is a study to evaluate feasibility/performance, safety and initial efficacy, and the sample size is small, no formal null hypotheses will be tested, and no imputation methods will be employed to replace missing or spurious data.

Numeric descriptive statistics will include the n (observed data), mean, standard deviation (SD) or standard error of the mean (SEM), median, minimum value, and maximum value. Categorical summaries will show the number and percent of subjects in the levels associated with the variable summarized and the analysis performed. Percentages will be calculated based on the total number of subjects with non-missing data for the summary. For all statistical tests a p-value of less than or equal to 0.05 on a two-sided test, or less than or equal to 0.025 on a one-sided test will be considered statistically significant, unless otherwise stated. All confidence intervals will be 95% confidence intervals unless otherwise stated.

<u>This application covers three further supplementary work packages (WP) summarised below:</u>

1. **WP1: Health economics modelling to inform uptake of the MAGNET system – led by Dr Melody Ni**

In the last 12 months, we will evaluate the MAGNET system's cost-effectiveness to facilitate its adoption by the NHS. This includes conducting a literature review to gather clinical and economic evidence and developing a cost-benefit model to assess its economic impact. We will also interview clinical experts to understand current and potential future pathways incorporating the MAGNET system. Additionally, a workshop will be organised with key stakeholders to develop a logic model that outlines the clinical pathway, requirements for implementation, and necessary data for demonstrating the technology's impact, while identifying barriers and facilitators. This process will guide our cost-benefit analysis, incorporating relevant economic and clinical outcomes within NHS care pathways. We will use NHS reference pricing and assess the overall budget impact on the NHS. The findings will be presented to stakeholders, including decision makers, PPIE groups, users, and domain experts, to guide further product development and commercial strategy.

Deliverable

Agreed user scenario and pathway to impact.

Agreed care pathways for decision making.

Cost benefit of the technology when used within the (English) NHS

Barriers and facilitators.

1. **WP2: PPIE – led by Ms Beata Szyldik (Patient representative and PPI Group Chair)**

We will conduct PPI activities in collaboration with NIHR HRC. The activities include:

1. Online feedback collection:

Use of social media platforms to access patient obesity forums including Facebook, where information about the potential trial will be disseminated and lay member's views sought.

2. Patient Advisory Group (PAG):

We will form a PAG by recruiting at least five patients/carers with relevant experience to support the project continuously. The PAG will convene quarterly online meetings, assist in designing Patient Information Sheets, optimise recruitment strategies, and help create and distribute dissemination materials, such as reports, posts, articles, and videos, to targeted patient groups online and offline.

3. Survey

Guided by the PAG, we will create a PATIENT SURVEY to distribute to bariatric clinic patients, the same demographic targeted for trial recruitment. The survey data will be analysed and included in our final report.

4. Professionals' views on acceptability of intervention (using magnets to create anastomoses)

We will survey 15 healthcare professionals (including bariatric physicians, endocrinologists, sleep physicians, gastroenterologists, dieticians, psychiatrists, and nurse practitioners) specialising in obesity treatment. They will receive an email containing a trial information sheet and be asked for their opinions on the technique and any concerns they may have about referring patients.

5. EDI

We will implement EDI strategies to reduce barriers for PPI members to participation, ensuring diverse and equitable involvement in research activities, including flexible formats and tailored support.

1. **WP3: Regulatory Approval– led by Dr Michel Gagner (industry co-applicant)**

The MAGNET system is a compression anastomosis device using magnet technology with FDA 510(k) clearance as a Class II medical device. The system's technical documentation is currently under review for CE marking by the manufacturer's EU-notified body. GT Metabolic Solutions Inc. plans to seek UK regulatory approval following the CE Marking, emphasizing the need for region-specific clinical evidence to support regulatory and reimbursement decisions in the UK. The device's use, feasibility, safety, and efficacy are also being evaluated in other regions through clinical studies in Australia (NCT06473831), Italy (NCT06613711), and Chile (NCT06613724), to align with local healthcare practices and patient populations.

Ethical Considerations:

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The risks and benefits of the device and alternatives will be addressed in the informed consent form and during the informed consent process, including allowing patients to ask questions. The patients have other options for an anastomosis, the conventional methods of incision and hand suturing or intestinal stapling, or to not seek treatment.

We are currently in the process of submitting our REC application to the HRA for approval.

Dissemination of Findings:

Findings will be presented nationally, internationally, in high impact publications and NIHR reports. We will engage with the public through our PPI group, charities and social media.

6. Intellectual Property and Commercialisation

The study, conducted by Imperial College in collaboration with GT Metabolic Solutions Inc., will use the company's background IP at no cost. All IP from the product and study will be owned by GT Metabolic Solutions Inc. Imperial College will disclose all inventions from the trial to the company, which in turn will offer Imperial College the first option to negotiate for an exclusive, worldwide, royalty-bearing license on reasonable terms.

GT Metabolic Solutions Inc understands that the study may not produce new commercially relevant IP but values the data for UK regulatory approval and commercialisation. The results are intended to be publishable under standard academic practices.

Imperial College's technology transfer arm, Imperial Innovations, will evaluate the commercial potential of any IP developed, although the likelihood of generating significant commercial IP is low. Imperial Innovations will lead the commercialisation efforts for any resultant IP.

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Uploads

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Pre-submission